

ASHE COUNTY AP, NC, USA

WMO: 723146

Lat: 36.432N

Lon: 81.419W

Elev: 969

StdP: 90.21

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 53892

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.5	-9.1	-22.5	0.6	-3.7	-19.2	0.8	-4.3	16.7	-1.4	15.2	-1.1	4.9	320	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	28.0	19.6	27.4	19.6	26.3	19.3	21.9	25.5	21.3	25.0	20.7	24.4	2.7	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	17.8	24.0	20.2	16.7	23.2	19.4	15.9	22.5	69.5	25.8	66.9	25.1	64.6	24.3	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	10.6	8.5	DB	-14.5	30.1	3.5	2.0	-17.0	31.6	-19.0	32.8	-21.0	33.9	-23.5	35.4	(4)
(5)			WB	-15.5	22.9	3.1	1.1	-17.7	23.6	-19.5	24.2	-21.2	24.8	-23.5	25.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	1.1	3.0	6.7	12.0	16.3	20.0	21.3	20.5	17.6	12.4	7.2	3.4		
	DBStd	8.34	6.06	5.56	5.53	4.89	3.71	2.71	2.40	2.36	3.17	4.37	5.05	5.33		
	HDD10.0	1006	281	202	132	36	4	0	0	0	1	27	112	212		
	HDD18.3	2662	536	428	362	194	81	14	5	8	49	189	334	462		
	CDD10.0	1673	4	7	28	94	199	299	349	327	228	100	28	8		
	CDD18.3	289	0	0	0	3	18	64	96	76	27	4	0	0		
	CDH23.3	1363	0	0	1	15	76	323	467	351	117	13	1	0		
	CDH26.7	168	0	0	0	0	4	50	64	43	7	1	0	0		
(14) Wind	WSAvg	2.6	3.9	3.7	3.4	3.2	2.4	2.0	1.8	1.3	1.4	2.4	2.8	3.2		
(15) Precipitation	PrecAvg	1275	104	79	127	97	106	127	123	119	110	86	94	102		
	PrecMax	1647	276	183	231	200	244	244	188	281	409	330	251	213		
	PrecMin	945	26	26	33	16	35	34	65	21	11	0	13	32		
	PrecStd	203	65	38	49	48	54	59	35	65	83	71	58	36		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.9	17.9	22.2	25.1	27.2	30.1	30.2	29.1	27.6	25.2	21.3	17.8		
		MCWB	11.3	12.1	13.1	14.3	17.2	19.5	20.2	20.8	18.5	18.2	13.7	12.1		
	2%	DB	13.8	15.3	18.9	23.0	25.3	27.8	27.9	27.7	26.2	22.6	18.0	15.3		
		MCWB	10.0	11.1	11.5	14.1	16.9	19.0	20.0	20.2	18.4	16.1	12.2	11.6		
	5%	DB	12.3	13.2	17.1	22.0	23.9	26.9	27.3	27.0	24.8	20.8	16.5	13.0		
		MCWB	9.1	9.3	10.6	13.5	16.4	18.7	20.1	20.2	18.2	14.7	11.5	9.7		
	10%	DB	9.9	11.4	14.9	19.7	22.6	25.1	26.1	25.8	22.9	18.8	14.8	11.9		
		MCWB	7.0	7.5	9.2	13.0	16.1	18.6	19.9	19.8	18.0	14.8	10.6	9.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	14.2	14.3	17.1	20.1	22.0	22.7	22.8	22.0	20.4	16.5	15.1		
		MCDB	15.0	16.2	19.3	21.0	24.5	25.9	26.3	26.6	24.3	23.1	18.1	16.3		
	2%	WB	11.4	12.5	13.0	15.8	19.0	21.0	21.8	21.9	20.5	18.6	14.6	12.8		
		MCDB	13.5	14.4	16.7	20.3	22.8	25.2	25.6	25.4	23.3	20.5	17.0	14.5		
	5%	WB	9.6	10.3	11.9	14.8	18.0	20.2	21.3	21.3	19.5	17.1	13.1	11.0		
		MCDB	11.3	12.7	14.9	19.2	21.7	24.4	25.0	24.9	22.3	18.9	15.0	12.4		
	10%	WB	7.3	8.2	10.7	13.8	17.2	19.5	20.7	20.6	18.8	15.5	11.7	9.0		
		MCDB	8.8	10.8	14.1	18.0	20.9	23.6	24.4	24.1	21.6	17.9	13.8	11.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	8.7	9.0	9.5	9.1	8.8	8.2	8.2	8.6	8.8	9.2	8.0		
		MCDBR	9.9	10.6	10.8	11.1	10.4	10.2	9.2	9.7	10.6	10.7	10.7	9.7		
	5% WB	MCWBR	7.4	7.5	6.3	5.5	4.9	4.4	4.0	4.4	4.9	5.6	6.7	7.1		
		MCWBR	9.1	10.0	9.2	9.5	9.1	8.9	8.2	8.3	8.8	8.1	8.3	8.7		
(40) Clear-Sky Solar Irradiance	taub	taub	0.278	0.284	0.308	0.352	0.389	0.419	0.451	0.450	0.358	0.320	0.288	0.274		
		taud	2.557	2.532	2.506	2.384	2.282	2.223	2.169	2.147	2.401	2.511	2.575	2.597		
	Ebn at Noon	Ebn at Noon	932	964	964	929	894	864	832	825	896	907	912	916		
		Edh at Noon	76	88	99	118	133	141	148	148	109	89	74	69		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.37	3.00	4.11	5.14	5.62	6.04	5.61	5.22	4.47	3.60	2.72	2.06		
	RadStd	RadStd	0.17	0.34	0.25	0.41	0.43	0.42	0.41	0.27	0.41	0.43	0.25	0.18		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (44 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+150	+91

Nomenclature: See separate page